

Retail Sales Analysis

Data Manipulation Project by Monyai R

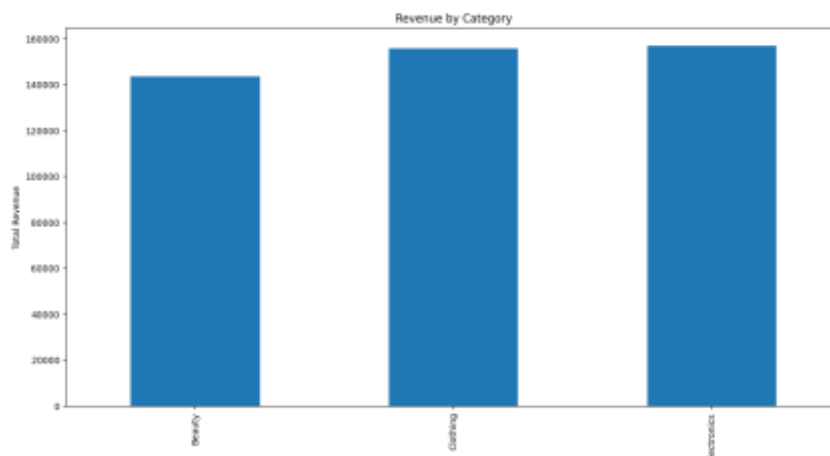
For this project I worked with a retail transactions dataset (1,000 rows) to practice the pandas skills I've been learning - grouping, multi-level indexing, handling missing values, and filtering. Instead of just running functions on the data to see what comes out, I set specific questions first and then used pandas to answer them. Below are the questions I asked and what I found.

What's in the dataset

Each row is a single transaction with a transaction ID, date, customer ID, gender, age, product category (Beauty, Clothing, or Electronics), quantity bought, price per unit, and total amount. Some quick stats: the average transaction is around R456, but the median is only R135, so most purchases are small and a handful of bigger purchases are pulling the average up. Average customer age is about 41, and the most expensive single transaction was R2,000.

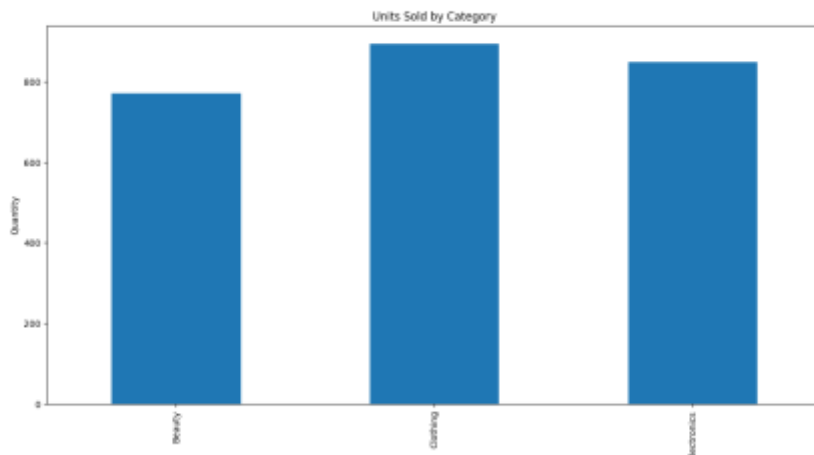
Which category makes the most money?

Electronics came out on top with about R157,000 in total revenue, just ahead of Clothing (R155,500) and Beauty (R143,500). It's a pretty close race between the three, no single category is really dominating.



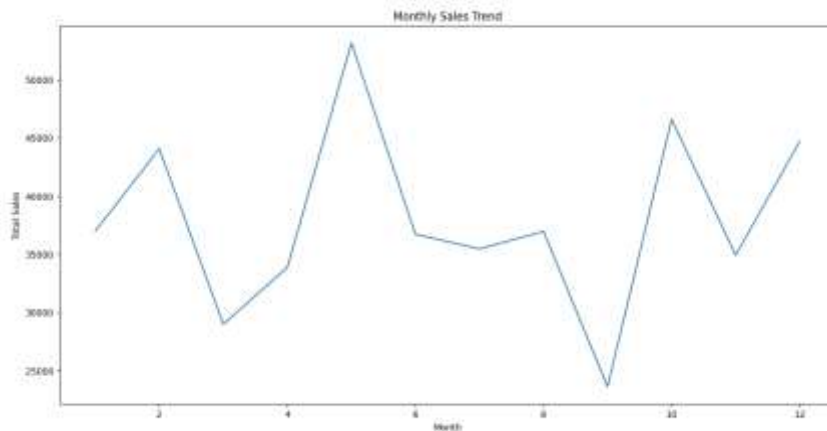
Which category sells the most units?

Here the order flips a bit - Clothing sells the most units (around 890), followed by Electronics (850) and Beauty (770). So Electronics makes more money without selling the most units, which makes sense once you check the price per unit - electronics items are priced higher on average, so fewer sales still add up to more revenue.



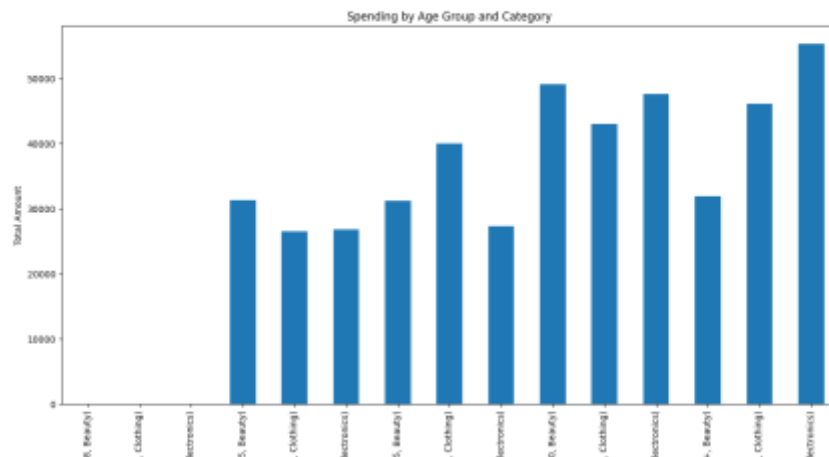
How do sales look month to month?

This one surprised me a bit. I expected something closer to a steady pattern (maybe higher near December), but instead sales bounce around a lot. May was the strongest month by far (about R53,000), and October had a smaller second peak (R46,500). September was the weakest month, at only around R23,700 - less than half of May. There's no obvious holiday-season pattern here, so whatever is driving the peaks and dips is probably something else, like promotions running in specific months, rather than regular seasonality.



Who spends the most - by age and category?

Older customers spend more overall, and this holds across all three categories, but it's most visible in Electronics - the oldest age group's electronics spending was clearly the highest bar on the chart. Younger customers (under 25 or so) contribute the least across the board. If this were a real business, older customers would probably be the group worth focusing retention or loyalty efforts on.



Takeaways

Putting it together: Electronics is the strongest category by revenue even though it doesn't sell the most units, because of its higher price point. Clothing moves the most volume. Sales aren't steady across the year - May and October stand out, September is a clear low point, and it doesn't line up with typical retail seasonality, so it's probably worth digging into what happened in those specific months (promotions, stock issues, etc.). And older customers are the highest-value segment, especially for electronics purchases.

This was a good exercise in asking a question first and then going to the data for the answer, instead of just generating charts and hoping something useful shows up.